PROBLEM K5 | Velocity potential | Polar coordinates | Acceleration | Unsteady

Oscillating Planar Dipole

Original statement · K5.pdf

OFFICIAL PROBLEM STATEMENT

For the planar fluid motion deriving from the velocity potential

$$\varphi = M\cos(\Omega t)\,\frac{\cos\theta}{r},$$

obtain:

1. Radial and azimuthal velocity components.
2. Expansion rate $\nabla \cdot v$.
3. Vorticity.
4. Acceleration of the fluid particle initially located at $r = 1$ and $\theta = \pi/2$.
5. Initial value of the circulation around a circle of radius $R = 1/2$ centered at $(r = 1, \theta = \pi/2)$.
6. Streamlines.
7. Trajectories and particle paths.

Source: K5.pdf (page 1). Notes.pdf, Chapter 2: gradient, divergence and curl in orthogonal coordinates Eqs. (2.3), (2.5), (2.6); potential Eq. (2.34); material derivative and acceleration Eqs. (2.23)-(2.27); circulation and Stokes theorem Eqs. (2.29)-(2.33); trajectories Eq. (2.12); stream lines Eq. (2.21).

Worked solution (4-phase method) expand / collapse

PHASE 1

Hypotheses, Degrees of Freedom and Data

- Planar, unsteady, potential flow $\mathbf{v} = \nabla\varphi$ in the plane (r, θ), for $r > 0$. The potential is that of a dipole whose strength $M\cos\Omega t$ oscillates in time.
- Degrees of freedom: two spatial coordinates (r, θ) plus time; the only time scale is Ω^{-1}.

- Dimensions of the constants follow from φ (a velocity times a length, m^2/s): $M\cos\Omega t/r$ has the units of φ, so M has units m^3/s.
- The origin is a singular point and is excluded.

Symbol	Meaning	SI unit
M	dipole strength (per unit depth)	m^3/s
Ω	angular frequency	s^{-1}
r, θ	polar coordinates (the data $r=1$, $R=1/2$ are lengths in m)	m, rad
φ	velocity potential	m^2/s

PHASE 2

Coordinates, Frames and Changes of Variable

- Inertial frame at rest with the dipole; polar coordinates (r,θ) with scale factors $(1, r)$, $x = r\cos\theta$, $y = r\sin\theta$, and $\mathbf{e}_z = \mathbf{e}_r \wedge \mathbf{e}_\theta$.
- Gradient: $\nabla\varphi = \partial_r\varphi\,\mathbf{e}_r + r^{-1}\partial_\theta\varphi\,\mathbf{e}_\theta$; divergence: $\nabla\cdot\mathbf{v} = r^{-1}\left[\partial_r(r v_r) + \partial_\theta v_\theta\right]$; vorticity: $\omega_z = r^{-1}\left[\partial_r(r v_\theta) - \partial_\theta v_r\right]$ (Eqs. 2.3, 2.5, 2.6).
- Acceleration in curvilinear coordinates: the component form $a_i = Dv_i/Dt$ of Eq. (2.27) holds only in Cartesian coordinates. The coordinate-free form of Eq. (2.26), $\mathbf{a} = \partial_t\mathbf{v} + \nabla\left(|\mathbf{v}|^2/2\right) - \mathbf{v}\wedge(\nabla\wedge\mathbf{v})$, is used instead.
- For part 7 the material coordinates are the initial position (r_0,θ_0) at $t=0$.

PHASE 3

Step-by-Step Derivation

3.1 Velocity components

Why this tool: since the motion derives from a potential, $\mathbf{v} = \nabla\varphi$ (Eq. 2.34), with the polar gradient above. Write $c(t) \equiv \cos\Omega t$.

$$v_r = \frac{\partial\varphi}{\partial r} = Mc\cos\theta\,\frac{\partial}{\partial r}\left(\frac{1}{r}\right) = \boxed{-\frac{Mc\cos\theta}{r^2},} \qquad v_\theta = \frac{1}{r}\frac{\partial\varphi}{\partial\theta} = \frac{1}{r}\cdot\frac{Mc}{r}\,(-\sin\theta) = \boxed{-\frac{Mc\sin\theta}{r^2}.}$$

3.2 Expansion rate

Why this tool: $\nabla \cdot \mathbf{v}$ is the rate of volume change per unit volume (Eqs. 2.62-2.63).

$$r\, v_r = -\frac{Mc\cos\theta}{r} \;\Rightarrow\; \frac{\partial(r v_r)}{\partial r} = \frac{Mc\cos\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial \theta} = -\frac{Mc\cos\theta}{r^2},$$

$$\nabla \cdot \mathbf{v} = \frac{1}{r}\left[\frac{Mc\cos\theta}{r^2} - \frac{Mc\cos\theta}{r^2}\right] = \boxed{0.}$$

The flow is incompressible for all t (consistent with $\nabla^2\varphi = 0$).

3.3 Vorticity

Why this tool: $\omega_z = r^{-1}\left[\partial_r(r v_\theta) - \partial_\theta v_r\right]$ is twice the angular velocity of the fluid element (Eq. 2.52).

$$r\, v_\theta = -\frac{Mc\sin\theta}{r} \;\Rightarrow\; \frac{\partial(r v_\theta)}{\partial r} = \frac{Mc\sin\theta}{r^2}, \qquad \frac{\partial v_r}{\partial \theta} = \frac{Mc\sin\theta}{r^2},$$

$$\omega_z = \frac{1}{r}\left[\frac{Mc\sin\theta}{r^2} - \frac{Mc\sin\theta}{r^2}\right] = \boxed{0.}$$

The motion is irrotational, as it must be for $\mathbf{v} = \nabla\varphi$ ($\nabla \wedge \nabla\varphi \equiv 0$).

3.4 Acceleration of the particle initially at $r = 1$, $\theta = \pi/2$

Why this tool: the acceleration is the material derivative of the velocity (Eq. 2.25). Because $\boldsymbol{\omega} = 0$, Eq. (2.26) reduces to $\mathbf{a} = \partial_t \mathbf{v} + \nabla(|\mathbf{v}|^2/2)$, which is valid in polar coordinates.

Local term, using $\partial_t c = -\Omega \sin\Omega t$:

$$\frac{\partial v_r}{\partial t} = \frac{M\Omega \sin\Omega t \cos\theta}{r^2}, \qquad \frac{\partial v_\theta}{\partial t} = \frac{M\Omega \sin\Omega t \sin\theta}{r^2}.$$

Convective term: $|\mathbf{v}|^2 = v_r^2 + v_\theta^2 = \dfrac{M^2c^2\left(\cos^2\theta + \sin^2\theta\right)}{r^4} = \dfrac{M^2c^2}{r^4}$, hence

$$\frac{|\mathbf{v}|^2}{2} = \frac{M^2c^2}{2r^4}, \qquad \frac{\partial}{\partial r}\left(\frac{|\mathbf{v}|^2}{2}\right) = -\frac{2M^2c^2}{r^5}, \qquad \frac{1}{r}\frac{\partial}{\partial\theta}\left(\frac{|\mathbf{v}|^2}{2}\right) = 0.$$

Field acceleration:

$$a_r = \frac{M\Omega\sin\Omega t\cos\theta}{r^2} - \frac{2M^2\cos^2\Omega t}{r^5}, \qquad a_\theta = \frac{M\Omega\sin\Omega t\sin\theta}{r^2}.$$

Evaluate at $r = 1, \theta = \pi/2$ ($\cos\theta = 0, \sin\theta = 1$):

$$\boxed{\mathbf{a}(1, \tfrac{\pi}{2}, t) = -2M^2\cos^2(\Omega t)\,\mathbf{e}_r + M\Omega\sin(\Omega t)\,\mathbf{e}_\theta.}$$

This is the acceleration of whichever particle occupies the point $(1, \pi/2)$ at time t. For the particle that is at that point at the initial instant, set $t = 0$: $\mathbf{a} = -2M^2\mathbf{e}_r$ (the local term vanishes because $\sin 0 = 0$). For $t > 0$ that particle has moved away from the point, so its acceleration is the field value at its new position (part 7).

3.5 Initial circulation around the circle of radius $1/2$ centred at $(r = 1, \theta = \pi/2)$

Why this tool: $\Gamma = \oint \mathbf{v}\cdot d\mathbf{l}$ (Eq. 2.29) equals the flux of the vorticity through any surface bounded by the curve that lies in the fluid (Eq. 2.32). The centre is at Cartesian $(0, 1)$ and the circle has radius $1/2$, so every point of the disc is at a distance of at least $1/2$ from the origin: the disc lies entirely inside the domain, where $\omega_z = 0$.

$$\Gamma = \iint_S \omega_z\, d\sigma = \iint_S 0\, d\sigma = \boxed{0}$$

at $t = 0$ and, since $\omega_z = 0$ for all t, at every instant. In fact φ is single-valued, so $\oint \nabla\varphi \cdot d\mathbf{l} = \varphi(\text{end}) - \varphi(\text{start}) = 0$ for every closed curve, including those that enclose the origin.

3.6 Streamlines

Why this tool: streamlines are tangent to $\mathbf{v}$ at a frozen instant (Eq. 2.21): $dr/v_r = r\,d\theta/v_\theta$. At an instant with $\cos\Omega t \neq 0$ the common factor $-Mc/r^2$ cancels:

$$\frac{dr}{\cos\theta} = \frac{r\,d\theta}{\sin\theta} \;\Longrightarrow\; \frac{dr}{r} = \frac{\cos\theta}{\sin\theta}\,d\theta \;\Longrightarrow\; \ln\frac{r}{r_0} = \ln\left|\frac{\sin\theta}{\sin\theta_0}\right|,$$

$$\boxed{r = \frac{r_0}{\sin\theta_0}\sin\theta = C\sin\theta.}$$

In Cartesian variables $r^2 = Cr\sin\theta$ gives $x^2 + y^2 = Cy$: circles of diameter C centred at $(0, C/2)$ and tangent to the x-axis at the origin. The shape does not depend on t. (At the isolated instants with $\cos\Omega t = 0$ the whole velocity field vanishes.) Equivalent form: the stream function $\psi = -Mc\sin\theta/r$ (with $v_r = r^{-1}\partial_\theta\psi$, $v_\theta = -\partial_r\psi$) has the level curves $\sin\theta/r = \text{const}$.

3.7 Trajectories and particle paths

Why this tool: trajectories solve $d\mathbf{x}/dt = \mathbf{v}$ (Eq. 2.12), here

$$\frac{dr}{dt} = -\frac{Mc\cos\theta}{r^2}, \qquad r\frac{d\theta}{dt} = -\frac{Mc\sin\theta}{r^2}, \qquad r(0) = r_0,\ \theta(0) = \theta_0.$$

Dividing the first equation by the second eliminates both t and the factor $c(t)$:

$$\frac{dr}{r\,d\theta} = \frac{\cos\theta}{\sin\theta} \;\Longrightarrow\; \boxed{r = r_0\frac{\sin\theta}{\sin\theta_0}}$$

which is the path line. It coincides with the streamlines of 3.6 because the direction of $\mathbf{v}$ does not depend on time (only its magnitude does).

Time law. Substitute the path into the second equation, $r^3 = r_0^3\sin^3\theta/\sin^3\theta_0$:

$$\frac{d\theta}{dt} = -\frac{Mc\sin\theta}{r^3} = -\frac{M\sin^3\theta_0}{r_0^3}\,\frac{\cos\Omega t}{\sin^2\theta} \;\Longrightarrow\; \sin^2\theta\,d\theta = -\frac{M\sin^3\theta_0}{r_0^3}\cos\Omega t\,dt.$$

Integrate with Barrow's rule between $(\theta_0, 0)$ and (θ, t). For the left side use $\sin^2\theta' = (1-\cos 2\theta')/2$, whose primitive is $\theta'/2 - \sin 2\theta'/4$; for the right side, $\int_0^t \cos\Omega t'\,dt' = \sin(\Omega t)/\Omega$:

$$\boxed{\frac{\theta-\theta_0}{2} - \frac{\sin 2\theta - \sin 2\theta_0}{4} = -\frac{M\sin^3\theta_0}{r_0^3\,\Omega}\sin\Omega t.}$$

Together with $r = r_0\sin\theta/\sin\theta_0$ this defines $\mathbf{x}(t)$ implicitly. The right-hand side vanishes whenever $\sin\Omega t = 0$, so the particle returns to its starting point every half period π/Ω: it oscillates back and forth along its circular path. (Particles on the axis $\theta_0 = 0$ or π stay on it, with $r^3 = r_0^3 \mp 3(M/\Omega)\sin\Omega t$.)

PHASE 4

Interpretation, Limits and Dimensional Check

- The flow is the oscillating field of a point dipole: incompressible, irrotational, with streamlines that are circles through the singular point and a velocity that decays as r^{-2}.
- Steady limit $\Omega \to 0$ (with $c \to 1$): the local term vanishes and the acceleration is purely convective, $a_r = -2M^2/r^5$ and $a_\theta = 0$, equal to $\nabla(|\mathbf{v}|^2/2)$; the particle then drifts monotonically along the circle.
- Consistency of the answers: part 5 ($\Gamma = 0$) follows from part 3 ($\omega_z = 0$); part 7 reproduces part 6 geometrically.
- Dimensions (SI): $[v] = [M/r^2] = (\mathrm{m^3/s})/\mathrm{m^2} = \mathrm{m/s}$; $[a] = [M^2/r^5] = (\mathrm{m^6/s^2})/\mathrm{m^5} = \mathrm{m/s^2}$ and $[M\Omega/r^2] = (\mathrm{m^3/s})(1/\mathrm{s})/\mathrm{m^2} = \mathrm{m/s^2}$; the combination $M/(r_0^3\Omega)$ in 3.7 is $(\mathrm{m^3/s})/(\mathrm{m^3/s}) = 1$, dimensionless as required for an angle.
- Numerical cross-check: symbolic differentiation of φ gives $\nabla\cdot\mathbf{v} = 0, \omega_z = 0$ and, from the full material derivative in polar components (including the $-v_\theta^2/r$ and $v_r v_\theta/r$ terms), the same a_r, a_θ as above, in particular $\mathbf{a}(1, \pi/2, t) = -2M^2\cos^2\Omega t\,\mathbf{e}_r + M\Omega\sin\Omega t\,\mathbf{e}_\theta$. RK4 integration of the particle ODE with $M = 0.8\ \mathrm{m^3/s}$, $\Omega = 1.7\ \mathrm{s^{-1}}$,

$r_0 = 1$ m, $\theta_0 = 0.45\pi$ up to $t = 1.3$ s gives $r = 0.846496$ m, equal to $r_0 \sin\theta / \sin\theta_0$ at the integrated angle, and the two sides of the boxed time law, -0.363902 and -0.363902.

READING OF THE OFFICIAL HANDWRITTEN SOLUTION

K5.pdf evaluates the acceleration at the field point $(1, \pi/2)$ for general t, obtaining the vector given in 3.4; the value for the particle that is there at $t = 0$ is its $t = 0$ value. The handwritten trajectory law agrees with the boxed relation in 3.7 and notes that streamlines and path lines coincide. No numerical discrepancy was found.

Study notes written from the official course material (Notes.pdf, Chapter 2 · p. 18). Online version: subjects/fluid-mechanics/problemas/topic-2-kinematics.html